1 I

1.1 (1)

Since the Lorentz force is the only force acting on the particle, we have:

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}.$$

In cylindrical coordinates, the equation of motion can be written as:

$$M\frac{d}{dt}\left(v_r\hat{\boldsymbol{r}} + v_\phi\hat{\boldsymbol{\phi}} + v_z\hat{\boldsymbol{z}}\right) = qB_z\left(v_\phi\hat{\boldsymbol{r}} - v_r\hat{\boldsymbol{\phi}}\right).$$

Since the radius of the helical motion is constant, we have $v_r = 0$. Noticing that

$$\frac{d\hat{\boldsymbol{\phi}}}{dt} = -\dot{\phi}\hat{\boldsymbol{r}}, \qquad \frac{d\hat{\boldsymbol{z}}}{dt} = 0,$$

the equation of motion becomes:

$$M\left(-v_\phi\dot{\phi}\hat{\boldsymbol{r}} + \dot{v}_\phi\hat{\boldsymbol{\phi}} + \dot{v}_z\hat{\boldsymbol{z}}\right) = qv_\phi B_z\hat{\boldsymbol{r}}.$$

Comparing the components, we obtain:

$$\begin{aligned} -Mv_\phi\dot{\phi} &= qv_\phi B_z, \\ \dot{v}_\phi &= 0, \\ \dot{v}_z &= 0. \end{aligned}$$

Using the relation $v_\phi = R\dot{\phi}$, we obtain:

$$-\frac{Mv_\phi^2}{R} = qv_\phi B_z.$$

Thus, the radius of the helical motion is:

$$\boxed{R = -\frac{Mv_\phi}{qB_z}}.$$

Since $R > 0$, $q > 0$, and $B_z > 0$, we must have:

$$\boxed{v_\phi < 0}.$$

1.2 (2)

The magnetic moment of the particle is defined as:

$$\boldsymbol{m} = \frac{1}{2}q\boldsymbol{r} \times \boldsymbol{v}.$$

Taking its z-component, we obtain:

$$m_z = \frac{1}{2}qRv_\phi.$$

Substituting the result of (1), we have:

$$\begin{aligned} m_z &= \frac{1}{2} q \left(-\frac{M v_\phi}{q B_z} \right) v_\phi \\ &= -\frac{M v_\phi^2}{2 B_z}. \end{aligned}$$

Therefore:

$$\boxed{m_z = -\frac{M v_\phi^2}{2 B_z}}.$$

2 II

2.1 (3)

The Lorentz force is given by:

$$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B},$$

where

$$\boldsymbol{v} = v_r \hat{\boldsymbol{r}} + v_\phi \hat{\boldsymbol{\phi}} + v_z \hat{\boldsymbol{z}}, \qquad \boldsymbol{B} = B_r \hat{\boldsymbol{r}} + B_z \hat{\boldsymbol{z}}.$$

Expanding the cross product, we obtain:

$$\boldsymbol{F} = q \left[v_\phi B_z \hat{\boldsymbol{r}} + (v_z B_r - v_r B_z) \hat{\boldsymbol{\phi}} - v_\phi B_r \hat{\boldsymbol{z}} \right].$$

Thus, the z-component of the Lorentz force is:

$$\boxed{F_z = -q v_\phi B_r}.$$

2.2 (4)

According to Gauss's law for magnetism:

$$\oint_S \boldsymbol{B} \cdot d\boldsymbol{S} = 0.$$

For a cylindrical surface of radius r and infinitesimal height dz, the total magnetic flux is:

$$\begin{aligned} 0 = {} & \left[B_z(z + dz) - B_z(z) \right] \pi r^2 \\ & + 2\pi r \int_z^{z+dz} B_r(r, z')\, dz'. \end{aligned}$$

Since dz is infinitesimal, we can approximate:

$$\begin{aligned} B_z(z + dz) - B_z(z) &\approx \frac{dB_z}{dz} dz, \\ \int_z^{z+dz} B_r(r, z')\, dz' &\approx B_r(r, z)\, dz. \end{aligned}$$

Substituting these approximations, we obtain:

$$\pi r^2 \frac{dB_z}{dz} dz + 2\pi r B_r\, dz = 0.$$

Therefore:

$$\boxed{B_r = -\frac{r}{2} \frac{dB_z}{dz}}.$$

2.3 (5)

According to (3), the z-component of the Lorentz force is:

$$F_z = -qv_\phi B_r.$$

Substituting the result of (4) and taking $r = R$, we obtain:

$$F_z = \frac{1}{2} qv_\phi R \frac{dB_z}{dz}.$$

Using the relation obtained in (1):

$$R = -\frac{Mv_\phi}{qB_z},$$

we finally obtain:

$$\begin{aligned} F_z &= \frac{1}{2} qv_\phi \left(-\frac{Mv_\phi}{qB_z} \right) \frac{dB_z}{dz} \\ &= -\frac{Mv_\phi^2}{2B_z} \frac{dB_z}{dz}. \end{aligned}$$

Thus:

$$\boxed{F_z = -\frac{Mv_\phi^2}{2B_z} \frac{dB_z}{dz}}.$$

2.4 (6)

According to (5), the equation of motion along the z-direction is:

$$M\frac{dv_z}{dt} = -\frac{M}{2} \frac{v_\phi^2}{B_z} \frac{dB_z}{dz}.$$

Using the chain rule:

$$\frac{dv_z}{dt} = \frac{dv_z}{dz} \frac{dz}{dt} = v_z \frac{dv_z}{dz},$$

we obtain:

$$v_z \frac{dv_z}{dz} = -\frac{v_\phi^2}{2B_z} \frac{dB_z}{dz}.$$

Since the Lorentz force does no work, the kinetic energy is conserved. Under the given approximation:

$$v_\phi^2 + v_z^2 = \text{constant}.$$

Differentiating with respect to z, we have:

$$v_\phi \frac{dv_\phi}{dz} + v_z \frac{dv_z}{dz} = 0.$$

Substituting the equation of motion, we obtain:

$$v_\phi \frac{dv_\phi}{dz} = \frac{v_\phi^2}{2B_z} \frac{dB_z}{dz}.$$

Rearranging:

$$\frac{2}{v_\phi}\frac{dv_\phi}{dz} = \frac{1}{B_z}\frac{dB_z}{dz}.$$

Integrating both sides with respect to z:

$$\ln(v_\phi^2) = \ln B_z + \text{constant}.$$

Therefore:

$$\ln\left(\frac{v_\phi^2}{B_z}\right) = \text{constant},$$

which gives:

$$\boxed{\frac{v_\phi^2}{B_z} = \text{constant}}.$$

3 III

3.1 (7)

According to (6), we have:

$$\frac{v_\phi^2}{B_z} = \frac{v_{\phi 0}^2}{B_0}.$$

Furthermore, from the conservation of kinetic energy:

$$v_\phi^2 + v_z^2 = v_{\phi 0}^2 + v_{z0}^2.$$

Therefore:

$$v_\phi^2 = v_{\phi 0}^2 + v_{z0}^2 - v_z^2.$$

Substituting this into the result of (6), we obtain:

$$\frac{v_{\phi 0}^2 + v_{z0}^2 - v_z^2}{B_z} = \frac{v_{\phi 0}^2}{B_0}.$$

At the reflection point, the velocity along the z-direction becomes zero:

$$v_z = 0.$$

Thus, the magnetic field at the reflection point satisfies:

$$\frac{v_{\phi 0}^2 + v_{z0}^2}{B_z} = \frac{v_{\phi 0}^2}{B_0}.$$

Rearranging, we finally obtain:

$$\boxed{B_z = \left[1 + \left(\frac{v_{z0}}{v_{\phi 0}}\right)^2\right] B_0}.$$

3.2 (8)

According to (1), the radius of the approximately circular motion is:

$$R = -\frac{Mv_\phi}{qB_z}.$$

Since M and q are constant, we have:

$$R^2 = \frac{M^2 v_\phi^2}{q^2 B_z^2}.$$

According to (6):

$$\frac{v_\phi^2}{B_z} = \text{constant}.$$

Therefore:

$$R^2 = \frac{M^2}{q^2 B_z}\left(\frac{v_\phi^2}{B_z}\right) \propto \frac{1}{B_z}.$$

Thus:

$$\boxed{R \propto \frac{1}{\sqrt{B_z}}}.$$

Since B_z increases as the particle approaches the reflection point, the radius of the circular motion decreases.

On the other hand, the magnetic flux through the circular orbit is:

$$\Phi = B_z \pi R^2.$$

Substituting the expression for R^2, we obtain:

$$\begin{aligned}\Phi &= \pi B_z \frac{M^2 v_\phi^2}{q^2 B_z^2} \\ &= \frac{\pi M^2}{q^2}\frac{v_\phi^2}{B_z}.\end{aligned}$$

Since v_ϕ^2/B_z is constant, we finally obtain:

$$\boxed{\Phi = \text{constant}}.$$

Therefore, as the particle approaches the reflection point, the radius of its circular orbit decreases, while the magnetic flux through the orbit remains constant.